

Ulam-ND

Assessment of mathematical legitimacy

Repository: [gewure/ulamnd](#) · Assessed revision: [a372a63](#) · 12 September 2026

Verdict: a substantive exploratory mathematics project, but not a reliable collection of proved results in its current form. The present manuscripts contain a false unsmoothed claim, an incorrect general explicit formula, and a significant defect in a spectral proof.

This assessment concerns the revision linked in reference [1]. It is not an allegation of fraud, a judgment about the author's motives, or a rejection of every result in the repository. It is a targeted assessment of whether the current mathematical claims can be relied on.

What is credible

The underlying mathematics is genuine: polynomial prime values, Bateman–Horn constants, pair singular series, Dedekind zeta functions, and modular spectral theory are established subjects. Goldston and Suriajaya's cited work really does connect singular-series averages to zeta zeros. [9]

There is also a checkable algebraic core. Independently written arithmetic checks verified the **finite-prime version** of paper I's main exact identity in 24 cases across four polynomials, with exact rational equality. This supports that algebraic identity; it does not establish the later infinite-product or asymptotic conclusions.

The repository discloses extensive AI assistance, records corrections, and rejects its original Fibonacci-dimension and quasicrystal suggestions rather than continuing to promote them. Those are positive signs of transparency. [7] [8]

What prevents endorsement

- **Paper I:** the claimed bounded error for the unsmoothed sum is impossible even for the classical linear polynomial. A short contradiction is given on page 2.
- **Paper II:** the general explicit formula assumes a pole is simple when it can be double. A cyclic cubic example requires an additional logarithm; see page 3.
- **Paper IV:** the proof replaces a restricted arithmetic sum with a complete modular-orbit sum, dropping a condition that matters; see page 4.

The main nonlinear smoothed conjecture remains open, and statements about actual prime counts require Hardy–Littlewood-type assumptions. The project is not a proof of the Riemann Hypothesis or of Bateman–Horn. Its own notes report that professional feedback is still being sought. [2] [6] [7]

Prepared by ChatGPT at the user's request. This AI-generated technical assessment is not an external peer-review report. Scope, verification, and clickable references appear on page 5.

FINDING 1 / DEFINITE FALSE CLAIM

The unsmoothed error cannot stay bounded

Source: paper I, Conjecture 1, specifically the formula labelled eq:conjsharp. [2]

The claim

The manuscript conjectures a Cesàro-smoothed asymptotic and then asserts, “consequently,” the unsmoothed estimate

$$\Sigma_f(H) := \sum_{1 \leq h \leq H} (S_f(h) - C(f)^2) = -\frac{1}{2}C(f) \log H + O_f(1).$$

Here $S_f(h)$ is the pair singular series and $C(f)$ is the Bateman–Horn constant. The issue is the bounded remainder $O_f(1)$, not the mere presence of a logarithmic main term.

A contradiction in the classical case

Take $f(t) = t$, which is included in the statement. Then $C(f) = 1$, and S_f is the classical prime-pair singular series. Define

$$R(H) = \sum_{1 \leq h \leq H} (S_t(h) - 1) + \frac{1}{2} \log H.$$

The claimed estimate says $R(H) = O(1)$. For integers $H \geq 2$, subtraction gives

$$S_t(H) - 1 = R(H) - R(H-1) - \frac{1}{2} \log \left(\frac{H}{H-1} \right).$$

The right-hand side is bounded if R is bounded. Thus the conjectured estimate would imply that **the singular series itself is bounded**.

But for even H its established formula is

$$S_t(H) = 2C_2 \prod_{p|H, p>2} \frac{p-1}{p-2}, \quad C_2 = \prod_{p>2} \left(1 - \frac{1}{(p-1)^2} \right) > 0.$$

This formula appears both in paper I and in Goldston–Suriyaya. [2] [9] Set $H = \prod_{p \leq y} p$, the product of all primes up to y . Then

$$\log S_t(H) - \log(2C_2) = \sum_{2 < p \leq y} \log \left(1 + \frac{1}{p-2} \right) \rightarrow \infty,$$

because the sum of reciprocals of primes diverges and the summands above are asymptotic to $\frac{1}{p}$. Hence $S_t(H)$ is unbounded, contradicting the necessary consequence of the asserted estimate.

What this does—and does not—show

Established: the unsmoothed $O(1)$ assertion is false as stated. This is an analytic contradiction, not a numerical objection or an appeal to authority.

Not established: that the Cesàro-smoothed conjecture is false. Smoothing can suppress large individual jumps, so the same contradiction does not apply to it.

The research notes describe replacing an earlier unsmoothed $o(1)$ remainder with $O(1)$. That correction does not resolve this obstruction. The repeated explanation that the sharp sum has only bounded fluctuations must also be revised. [7]

FINDING 2 / INCORRECT GENERALIZATION

A higher-order pole changes the formula

Source: paper II, the theorem headed “general f ” and its proof, labelled `thm:explicit-general`. [3]

The missing possibility

The stated general explicit formula includes a contribution

$$c'_{f,m} x^{m-\frac{2}{3}}$$

from the pole at $s = -\frac{2}{3}$. That is the form of a **simple-pole** contribution. The manuscript’s own character construction allows the pole to have higher multiplicity; a double pole introduces a factor involving $\log x$.

A concrete polynomial

Consider

$$f(t) = t^3 - 3t - 1.$$

A monic cubic with constant term -1 can have a rational root only at 1 or -1 . Here $f(1) = -3$ and $f(-1) = 1$, so the polynomial is irreducible. Its discriminant is

$$-4(-3)^3 - 27(-1)^2 = 81.$$

An irreducible cubic with square discriminant has Galois group C_3 . The polynomial also has no fixed prime divisor, since $f(0) = -1$. Thus it is a natural example within the claimed general setting.

Apply the paper’s character formula

Let V be the permutation representation on the three roots. Its character has values $\chi_V = (3, 0, 0)$ on the three elements of C_3 . Every group element satisfies $g^3 = e$. The manuscript defines

$$\Psi_3(g) = \frac{\chi_V(g)^3 - \chi_V(g^3)}{3}.$$

Therefore

$$\Psi_3 = (8, -1, -1), \quad \langle \Psi_3, \mathbf{1} \rangle = \frac{8 - 1 - 1}{3} = 2.$$

The trivial character occurs twice. More explicitly, if χ and $\bar{\chi}$ are the two nontrivial characters of C_3 , then

$$\Psi_3 = 2\mathbf{1} + 3\chi + 3\bar{\chi}.$$

Consequently $L(3s + 3, \Psi_3)$ contains $\zeta(3s + 3)^2$, giving a double pole at $s = -\frac{2}{3}$. Under the theorem’s intended Artin/RH hypotheses, the other relevant factors do not cancel it in this example.

The required residue shape

If the Mellin integrand has Laurent expansion $\frac{a_{-2}}{(s-s_0)^2} + \frac{a_{-1}}{s-s_0} + \dots$ at $s_0 = -\frac{2}{3}$, then its product with x^{s+m} has residue

$$x^{m-\frac{2}{3}}(a_{-2} \log x + a_{-1}), \quad a_{-2} \neq 0.$$

Thus a constant coefficient $c'_{f,m}$ is insufficient. The general theorem must allow the polynomial in $\log x$ dictated by pole multiplicity.

Scope: this is a concrete defect in the general statement. It does not, by itself, refute the quadratic special case, whose corresponding pole is simple.

FINDING 3 / SIGNIFICANT PROOF DEFECT

A restriction disappears in the orbit argument

Sources: paper IV's model definition, the Heegner-orbit passage, and the final spectral theorem. [5] [4] [10]

The arithmetic object actually defined

The simplified model retains a coprimality condition:

$$\sigma_{-1}^*(n) = \sum_{d|n, \gcd(d, 2D)=1} \frac{1}{d},$$

$$\mathcal{G}(Y) = \sum_{h \leq Y} (Y - h)(\sigma_{-1}^*(h^2 - D) - \mathbb{E}\sigma_{-1}^*).$$

The author drops the multiplicative weight and squarefree condition, but **does not drop** $\gcd(d, 2D) = 1$. The manuscript explicitly distinguishes this model from the weighted pieces needed in paper III.

Where the proof changes the sum

In the Heegner parametrization, a quadratic form $[d, b, c]$ of discriminant D corresponds to

$$z = \frac{-b + \sqrt{D}}{2d}, \quad \Im z = \frac{\sqrt{|D|}}{2d}.$$

The proof's Step 2 then writes the sum over all Heegner points in complete level-one modular orbits. Step 3 unfolds a full level-one Poincaré series. These steps do not retain the condition $\gcd(d, 2D) = 1$.

That omission is not merely a choice of notation: the admitted subset is not a complete modular orbit.

An explicit orbit crossing

For $D = -4$, the form $[5, 4, 1]$ gives $z_5 = \frac{-2+i}{5}$, whose modulus is admitted since $\gcd(5, 8) = 1$. The determinant-one modular transformation below sends it to an excluded modulus:

$$\gamma = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \quad \gamma z_5 = \frac{z_5}{z_5 + 1} = \frac{-1 + i}{2} = z_2.$$

The latter corresponds to $[2, 2, 1]$, again of discriminant -4 , but $\gcd(2, 8) = 2$. Its modulus is excluded. Thus a single modular orbit contains both admitted and excluded points. This example uses $d = 5$ and $d = 2$, so it does not depend on any convention about the constant divisor $d = 1$.

Consequence for the theorem

Established: the complete-orbit identity does not describe the restricted arithmetic sum as written. The proof requires an explicit treatment of the coprimality restriction before the full-orbit spectral expansion can be used.

Not established: that no corrected spectral theorem is possible. This finding invalidates a step of the presented argument, not every potential repair.

Separately, paper IV openly states that its $u = 1$ result does not provide the uniform estimates over $u > 1$ needed for the full conjecture. That acknowledged open problem is **not** being counted here as a false theorem. The restriction defect above already arises at $u = 1$. [6]

How to use this assessment

What was independently checked

- **Exact identity:** the finite-prime version of paper I's identity was evaluated using rational arithmetic for $f = t, t^2 + 1, t^2 + t + 41$, and $t^3 - 3t - 1$; primes 2, 3, 5, 7, 11; and $H = 1, 2, 5, 10, 25, 100$. All 24 equalities held exactly.
- **Finding 1:** the contradiction follows from the established singular-series formula and divergence of prime reciprocals. Numerical evaluation of the primordial factors was supplementary, not the proof.
- **Finding 2:** the rational-root and discriminant calculations, character values, and trivial-character multiplicity were recomputed independently.
- **Finding 3:** the modular transformation and coprimality conditions were checked directly; both associated quadratic forms have discriminant -4 .

These checks were written independently rather than obtained by running the repository's own test suite. Agreement in finite arithmetic does not certify analytic continuation, error estimates, or exchanges of infinite sums.

Limits and recommendation

This is a targeted review, not a full referee report or a complete literature/novelty audit. The full numerical studies were not reproduced, and software execution or supply-chain security was not audited. No conclusion about fraud, motives, or the value of every unpublished idea follows from these findings.

Recommendation: treat the repository as exploratory research with potentially useful identities and numerical observations. Do not cite its new theorems as established or endorse the present manuscripts without substantial corrections and specialist review. AI involvement is not the basis for this judgment; the mathematical counterexample and proof defects are.

Clickable sources

All GitHub links below are pinned to revision a372a63c5be42d88905c6f6335582c8720bc8d80.

- [1] **Repository revision.** Commit inspected for this assessment.
- [2] **Paper I, main.tex.** Conjecture 1; classical singular series; exact identity; conditionality and AI disclosure.
- [3] **Paper II, main.tex, lines 579–605.** General explicit formula and character-multiplicity argument. The earlier character definition is in the same file.
- [4] **Paper IV, main.tex, lines 427–493.** Heegner parametrization, orbit identity, and spectral unfolding.
- [5] **Paper IV, main.tex, lines 931–943.** Model divisor sum retaining the condition $(d, 2D) = 1$.
- [6] **Paper IV, main.tex, lines 1050–1097.** Acknowledged lack of the required uniformity in u .
- [7] **research/KNOWLEDGE.md.** Submission status, research history, errors, and corrections. Historical status labels are not treated as proof.
- [8] **research/experiments/CONCLUSIONS.md.** Original experimental claims, negative findings, and stated numerical limitations.
- [9] **D. A. Goldston and A. I. Suriajaya.** "A singular series average and the zeros of the Riemann zeta-function," arXiv:2007.16099; published in *Acta Arithmetica* 200 (2021), 71–90.
- [10] **Paper IV, main.tex, lines 770–805.** Final spectral theorem and proof for the simplified model.